ENGINEERING PORTFOLIO · STRUCTURES & DESIGN

Nathaniel Murcin

Structural and fluid systems design, from concept to hardware.

Georgia Institute of Technology
B.S. Aerospace Engineering, expected May 2027
M.S. Aerospace Engineering, expected May 2028

BLUE ORIGIN · GTXR · PURSUIT AEROSPACE

nmurcin3@gatech.edu · nmurcin.github.io



Designing the fins

Fin Design & Manufacturing Lead, Georgia Tech Experimental Rocketry · Aug 2023–present

PROBLEM

- Early fin architecture needed two decisions: cross-section geometry and core material

CRITERIA

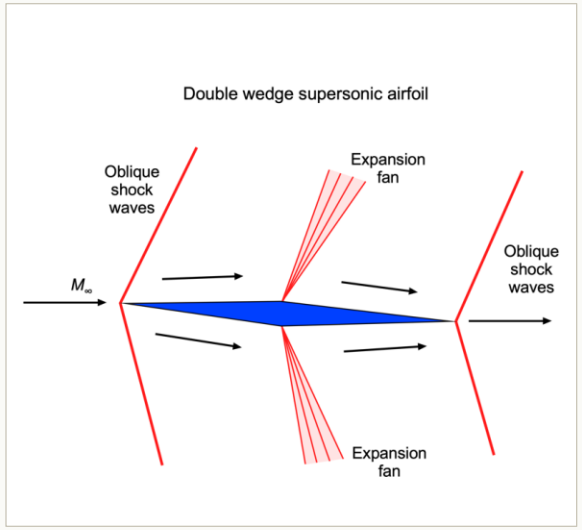
- Manufacturability & repeatability
- Mass
- Aerodynamic performance
- Process safety

DECISION

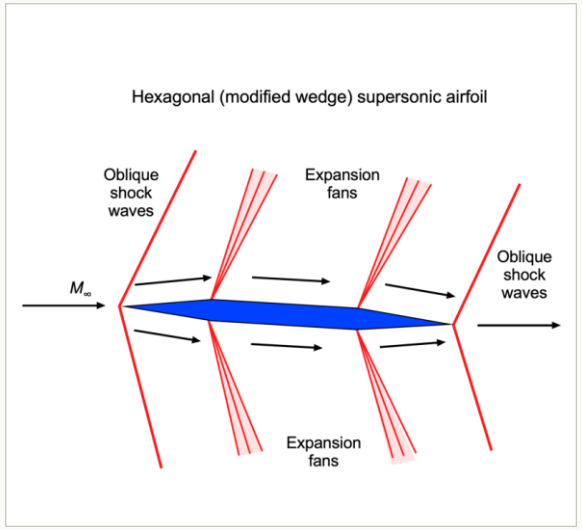
- Goal: an architecture that can be **built consistently across eight flight fins**, rather than one that maximizes single-fin performance

VERIFICATION

- Simulation results informed the architecture decisions
- Ownership boundary: detailed geometry optimization was done later by the GTXR simulations team



Double wedge
Aerodynamically preferred; harder to machine repeatably



Hexagonal
Repeatable to manufacture; higher predicted flutter margin

Shock diagrams: Introduction to Aerospace Flight Vehicles, Embry-Riddle Aeronautical University.

Decision tree trade study

PROBLEM

- Choose the fin cross-section and taper

CRITERIA

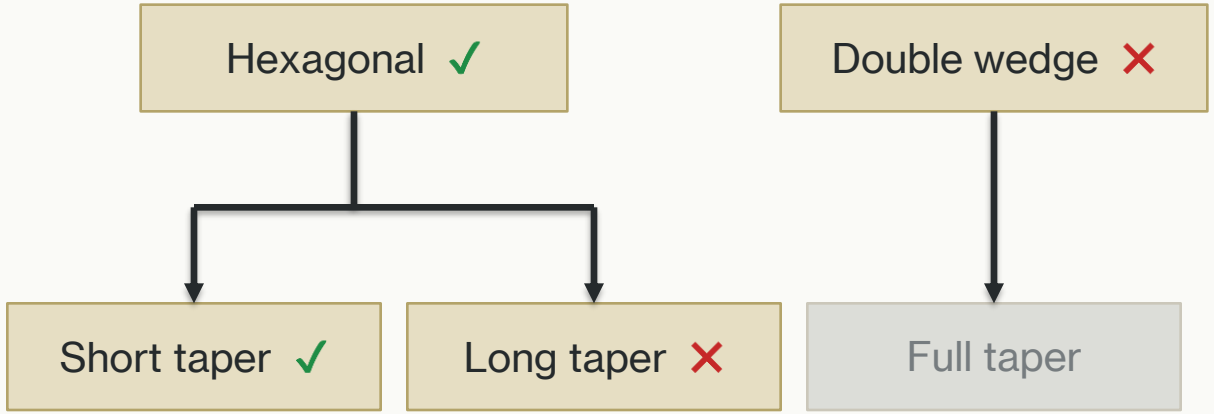
- Double wedge was aerodynamically preferred
- The alternative geometry carried only a small predicted apogee penalty
- Manufacturing repeatability became the primary discriminator

DECISION

- **Hexagonal**: repeatable to manufacture, higher predicted flutter margin; modest aerodynamic penalty accepted
- **Short taper**: easier machining and higher flutter margin
- Long taper rejected for machining difficulty; full taper dropped with the double wedge

VERIFICATION

- Simulation predictions: higher flutter margin, small apogee penalty



Detailed geometry optimization was performed later by the GTXR simulations team.

Design for reliability & feasibility

PROBLEM

- Select the core for the short-taper fin

CRITERIA

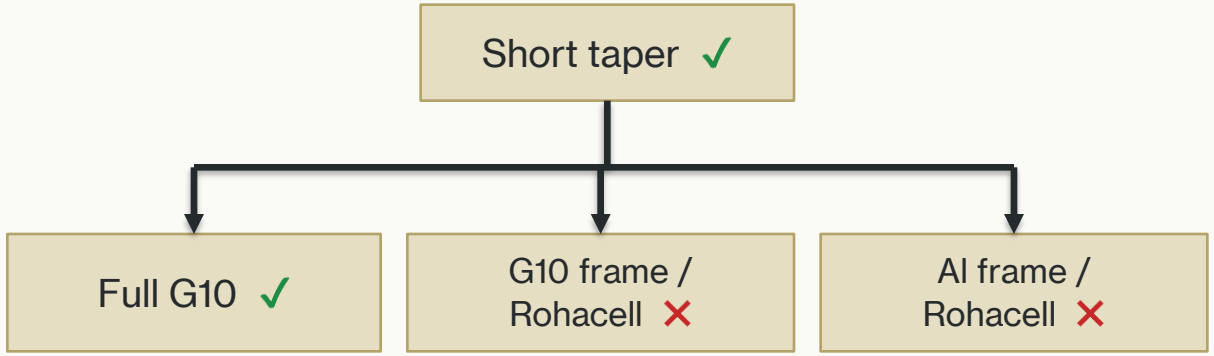
- Manufacturing confidence and repeatability across eight fins
- Mass
- Compatibility with the carbon-fiber skins

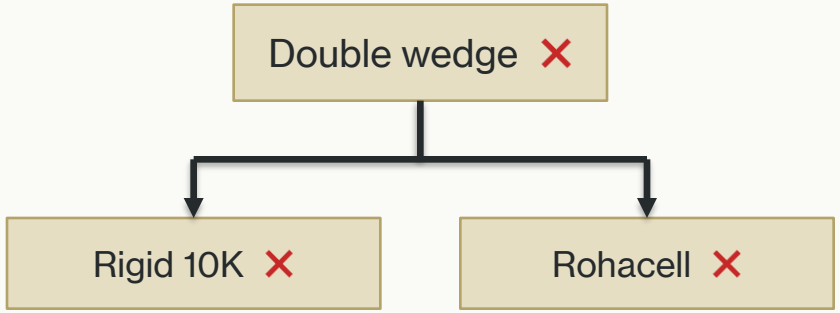
DECISION

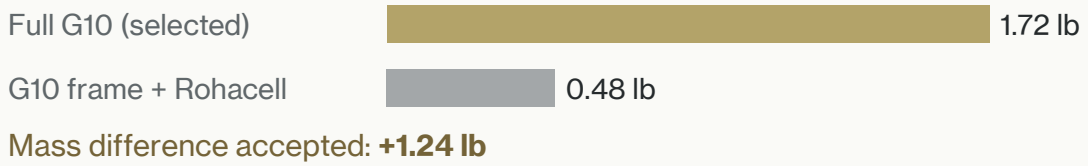
- ✕ G10 frame + Rohacell: 0.48 lb, lightest, but thermoforming was not yet mature or repeatable
- ✕ Al frame + Rohacell: CTE mismatch with carbon fiber; prior interface delamination
- ✕ Double-wedge cores: warping and immature processes
- ✓ **Full G10: 1.72 lb**, highest manufacturing confidence and repeatability

VERIFICATION

- +1.24 lb accepted for flutter margin, consistency and repeatability
- Follow-on: thermoformed Rohacell cores researched as a lighter alternative (AIAA 2026)







Designing a precision fin alignment jig

PROBLEM

- Four fins bonded at precise 90° spacing and 1.6° cant
- Alignment had to hold through bonding and cure
- Previous method: labor-intensive and hard to repeat

CRITERIA

- Clocking 90° ± 1°, from vehicle sensitivity analysis
- Cant 1.6° ± 0.1°
- Adapts to multiple rocket diameters and fin geometries
- Less setup and attachment time

DECISION

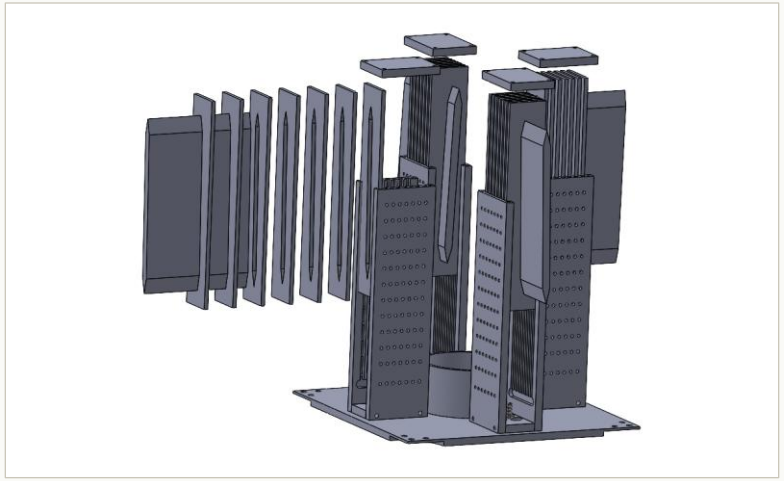
- Modular **four-tower fixture** designed around the fin-can geometry in SolidWorks
- All four fins positioned and bonded in one setup
- Led fixture design and the operating procedure; aided CAM programming and machining

VERIFICATION

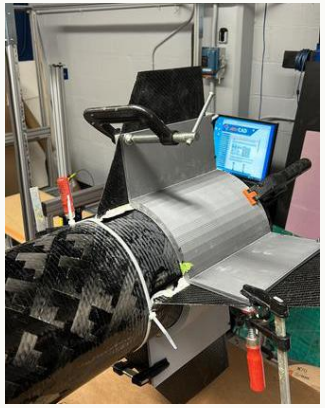
- ~1.5 hr labor saved; 24 hr cure-time reduction

NEXT

- Limitations identified and improvements designed; remachining planned for the next build cycle



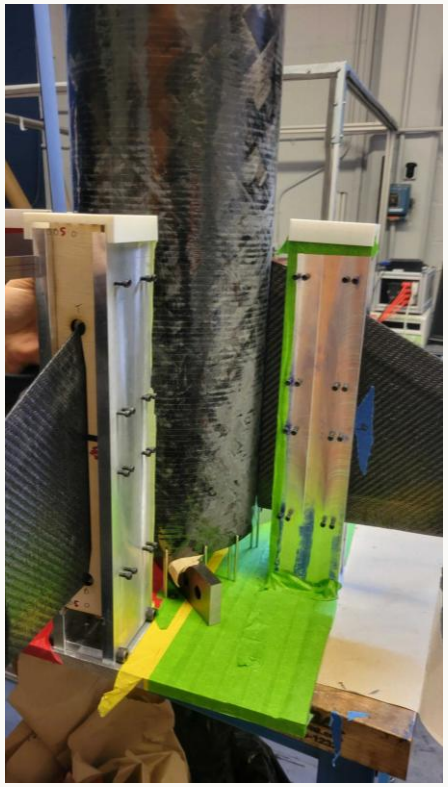
SolidWorks model: modular towers and base



Previous setup



Bonded fin with 1.6° cant



Fixture holding fins through bonding

Metrological 3D scan before jig

PROBLEM

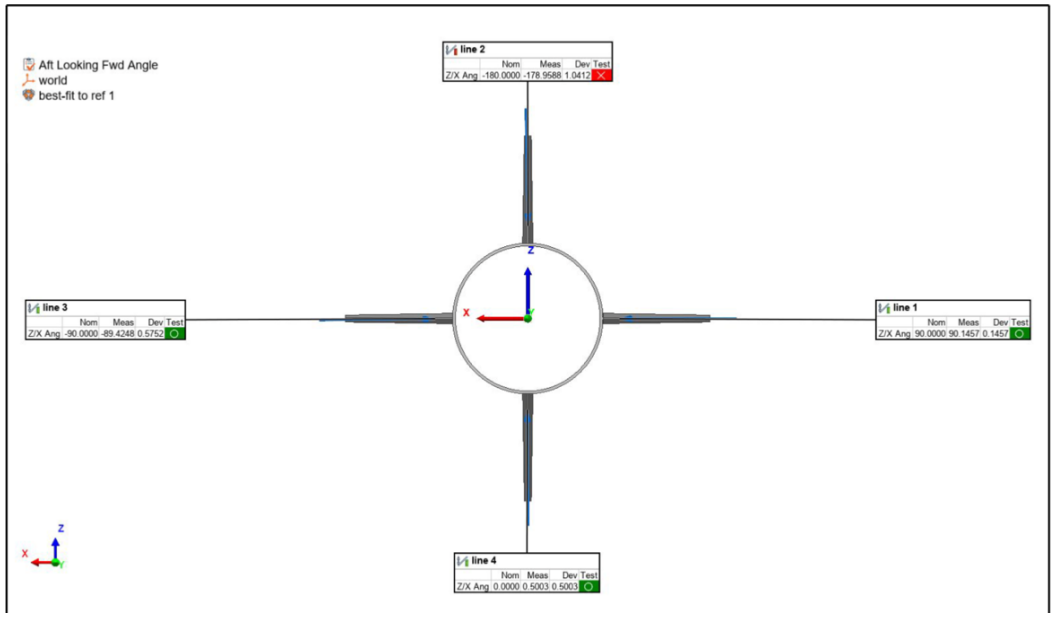
- Mini fin-can practice article, bonded with the previous method before the fixture existed
- 3D scan vs. nominal: 1 of 4 fins outside the ±1° clocking tolerance, at **1.041°**

CRITERIA

- Clocking 90° ± 1°, from vehicle sensitivity analysis
- Visual inspection alone could not verify alignment

DECISION

- Built the four-tower fixture; iterated the process and its usage instructions
- Standardized the layup and attachment sequence
- Improved layup application training
- Formal **3D-scan QC checkpoint** before flight hardware



Control View

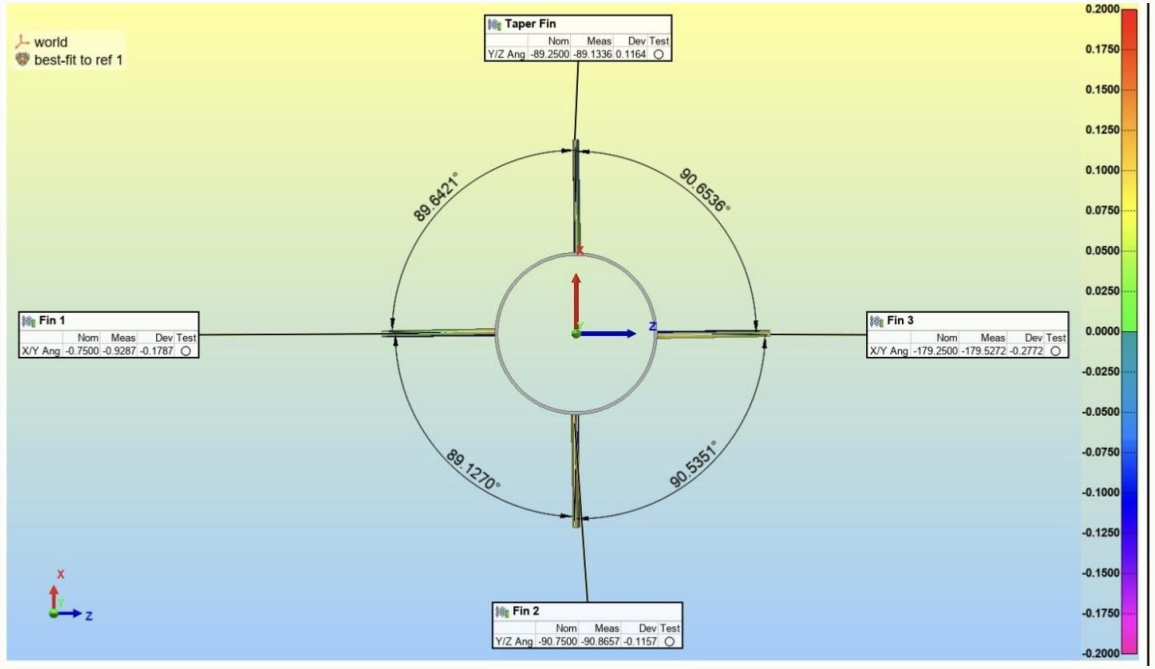
Control View Name	Aft Looking Fwd Angle
Units	Inches
Coordinate Systems	world
Data Alignments	best-fit to ref 1
All Statistics	Total: 4, Measured: 4 (100.0000%), Pass: 3 (75.0000%), Fail: 1 (25.0000%), Warning: 0 (0.0000%)

Char No.	Object Name	Control	Nom	Meas	Tol	Dev	Test	Out Tol
	line 1	Z/X Angle	90.0000	90.1457	±1.0000	0.1457	Pass	
	line 2	Z/X Angle	-180.0000	-178.9588	±1.0000	1.0412	Fail	0.0412
	line 3	Z/X Angle	-90.0000	-89.4248	±1.0000	0.5752	Pass	
	line 4	Z/X Angle	0.0000	0.5003	±1.0000	0.5003	Pass	

Practice article (mini fin-can) before the fixture: line 2 deviates 1.041° against a ±1° tolerance

Metrological 3D scan after jig

REQUIREMENT 1 · CLOCKING 90° ± 1°

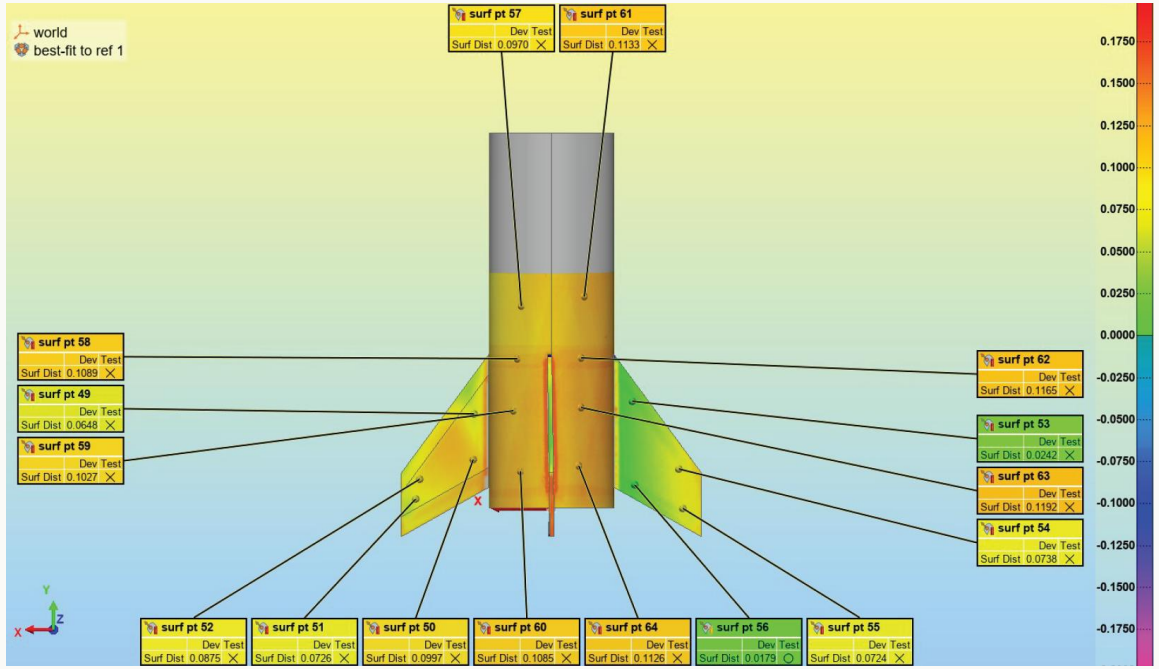


Practice article re-scan with the fixture (Fin 3 worst case: 0.277°)

RESULT

- Practice article: worst case 1.041° → 0.277° (~73% lower)
- Flight fins: clocking error **< 0.1°**

REQUIREMENT 2 · CANT 1.6° ± 0.1°



Practice article with the fixture: surface deviation vs. CAD (Hexagon)

RESULT

- Flight fins: all four within **0.1°** of the 1.6° cant target
- Fixture also saved ~1.5 hr of labor and 24 hr of cure time

Flight fins are not pictured: scans shown are of the practice article, made before the flight fins. The flight fins were scanned separately; results are from those scans.

Flight results

VERIFICATION

- Fin design cycle carried from trade study through build and inspection to flight hardware
- **Booster recovered** with fins intact in both vehicle cycles
- Sustainer had unrelated, non-structural anomalies, so its fins were not recovered



Recovered Mach N' Roll booster, fins intact



Mach N' Roll (2025)



Live and Let Fly (2024)

Plate-twist rig: validating shear modulus

I designed the plate-twist test apparatus · Co-author & presenter, AIAA 2024-85690 (7 authors)

PROBLEM

- Fin flutter predictions need the in-plane shear modulus of carbon-fiber/Rohacell sandwich panels
- Simulation-only values are hard to trust: process, bond quality, defects

CRITERIA

- Rigid plates sized to the 5 × 5 in panel; load points equidistant from center
- No local puncture of the panel
- Results within 5% of an ANSYS ACP model built by others

DECISION

- 1 in aluminum plates with ball-bearing contacts instead of puncture-prone cones
- Lower fixture rotated 90° to load each in-plane axis on an Instron 5982

VERIFICATION

- Three panels: 3.396 / 3.386 / 3.474 GPa vs. 3.281 GPa model
- **Mean within 5% of FEA** (≈ 4.2%)
- The validated value informed GTXR's fin optimizer

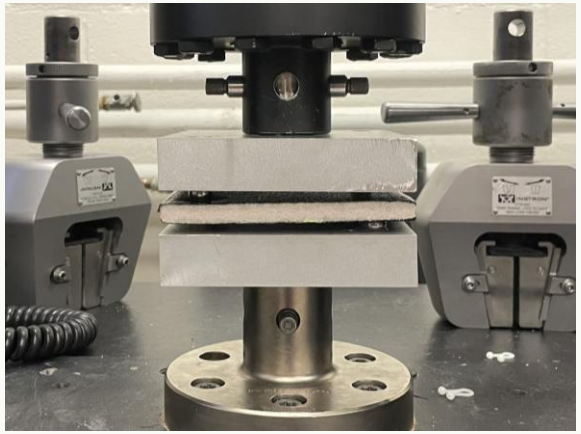
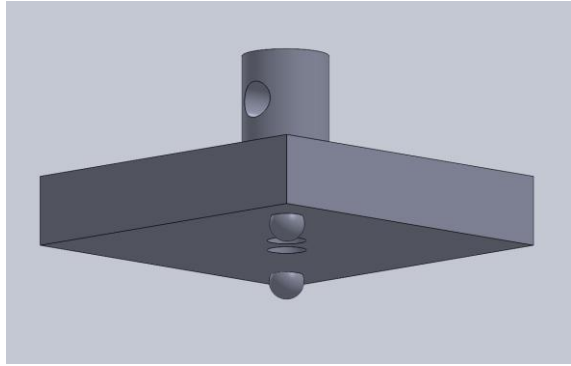
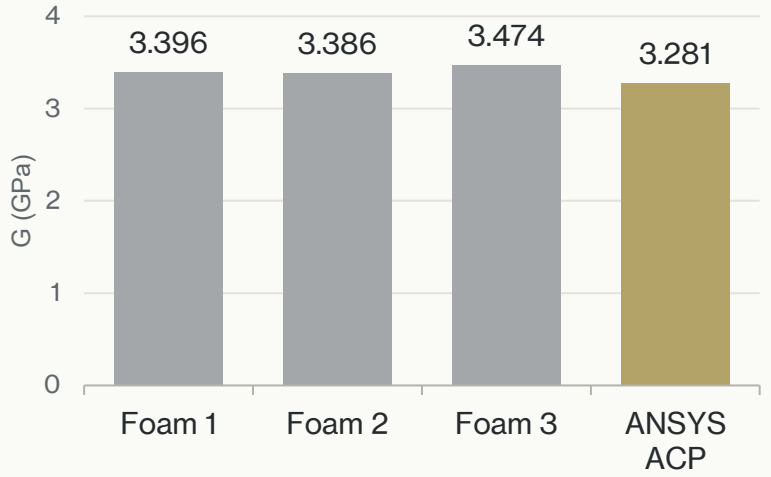


Plate-twist test on the Instron 5982



Rig CAD: ball-bearing contacts



≈ 4.2%

mean of three panels above the model

Lunar tooling structural analysis

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Grip lunar hardware without part slippage or damage

CRITERIA

- Positive margins against yield and ultimate for every load case

DECISION

- Analyzed the fixture **like a flight structure**
- Hand calculations for each load case, cross-checked with Creo FEA

VERIFICATION

- Each load case evaluated against the yield and ultimate margin criteria

STRUCTURAL ANALYSES PERFORMED

Torque resistance	Bolt shear & tension
Preload	Joint slip
Tear-out	Bending
Thermal expansion (CTE)	Creo FEA cross-check

No Blue Origin hardware images or specifications are shown.

Flight-critical design and build in 14 hours

Second-stage ignition passthrough, designed and built with the GTXR fin team

PROBLEM

- A flight-unit passthrough failed
- Second-stage ignition needed an electrical path through a pressure boundary

CRITERIA

- Seal at the NPT pressure-boundary interface
- Carry the electrical connection for second-stage ignition
- Pass a hydrostatic proof test at 2× MEOP before flight

DECISION

- Designed and fabricated a replacement flight-critical NPT passthrough with the fin team in **14 hours**

VERIFICATION

- Hydrostatic proof test at 2× MEOP, run by the propulsion lead
- Hardware flew; **the second stage ignited**
- The design informed later passthroughs



Finished NPT passthroughs

14 hr

failure to finished flight hardware

2× MEOP

hydrostatic proof (run by the propulsion lead)

Flew

second stage ignited

Mock launch rail improvements

PROBLEM

- The V1 rail flexed, was not straight, and did not hold the rocket steady

CRITERIA

- Stiff in bending and stable against buckling
- Straight and aligned while raising and holding the vehicle

DECISION

- With the structures lead, recommended an **I-beam rail** and a **larger base**
- Aided the design and load-path analysis
- Stay cables for alignment and raising

VERIFICATION

- Bending and buckling checks guided the design
- V2 in use for all-up practice



V1: rail flexed and drifted off straight



V2 during all-up practice

Thermoformed foam cores

Co-developed the process and machined the tooling · Co-author & presenter, AIAA 2026-112793 (3 authors)

PROBLEM

- Find a lighter alternative to dense fin cores: thermoform ROHACELL 51 into double-wedge cores

CRITERIA

- Geometric accuracy vs. CAD and repeatability across cores, measured by 3D scan

DECISION

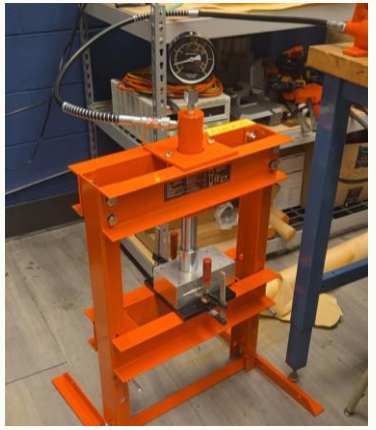
- Aluminum mold; blank heated to ~195 °C
- 4 t press load at the cavity centroid, not the geometric center, for planar contact
- Verdict: **promising, not yet flight-ready**; fixes proposed (mold borders / cutting inset, oversized blank)

VERIFICATION

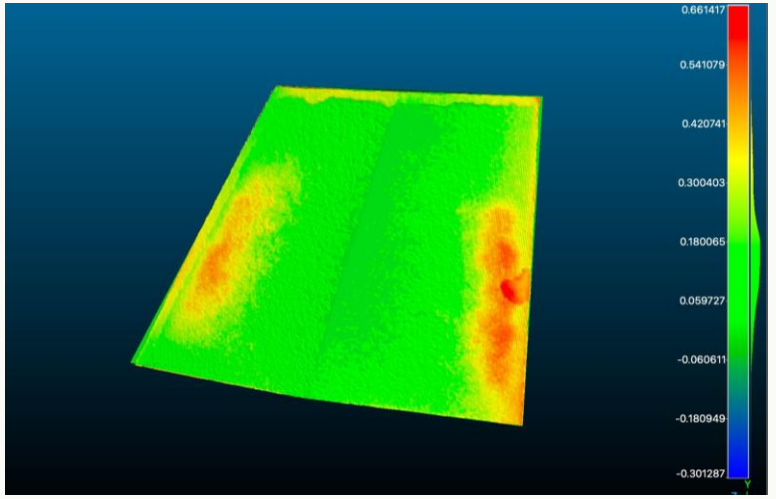
- 5 cores: 0.28 mm average global geometric error; 0.17 mm average surface std. dev.
- Error sources: mold thermal growth ≈ 0.59 mm (same order as the error), blank alignment, moisture, spring-back



Aluminum molds



Centroid-loaded press



Scan vs. CAD deviation, one formed core

0.28 mm
average global error, 5 cores

0.17 mm
average surface standard deviation

≈ 0.59 mm
mold thermal growth at forming temperature

Cryogenic tank-farm P&ID design

Blue Origin · Fluids & Manufacturing Engineering Intern · May–Aug 2026 · Cape Canaveral, FL

PROBLEM

- Design a critical-path cryogenic tank-farm P&ID
- Starting point: heritage designs

CRITERIA

- Meet the facility's requirements with the fewest components
- Every line, valve and relief device sized and validated
- Hazards analyzed before formal HAZOP review

DECISION

- Challenged requirements, eliminated branches and downsized hardware
- Sized, selected and validated cryogenic piping, valves and relief devices: Cv, Darcy-Weisbach, line sizing, overpressure relief
- Developed the BOM and coordinated suppliers

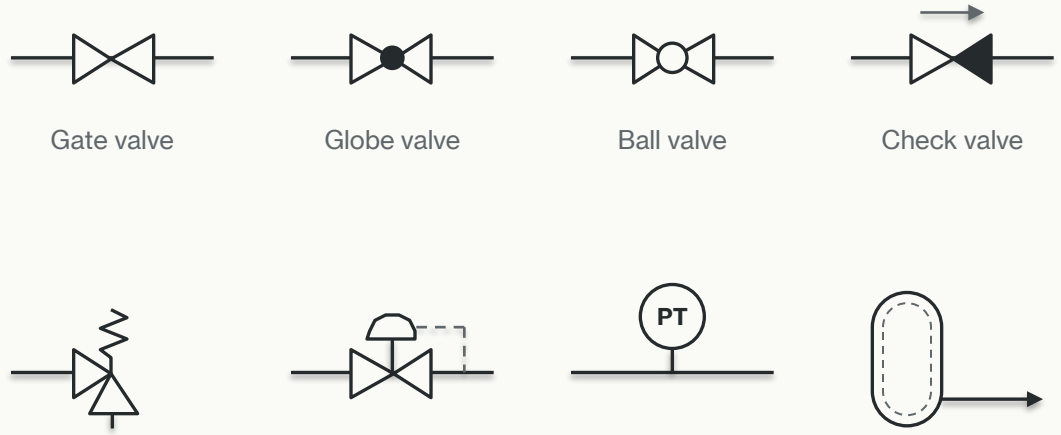
VERIFICATION

- Preliminary hazard analysis: component failure modes and consequences against safeguards
- Documented the CONOPS for formal HAZOP review

~60%

fewer components than heritage designs

P&ID SYMBOLS (GENERIC)



Generic P&ID symbols for illustration, not the Blue Origin system. No Blue Origin hardware, facilities, drawings or specifications are shown.

Fixtures and gauges cut production times

Engineering Intern · Aug–Dec 2025 · Newburyport, MA

PROBLEM

- Marking load/unload took 6 s per part; inspection took 40 s

CRITERIA

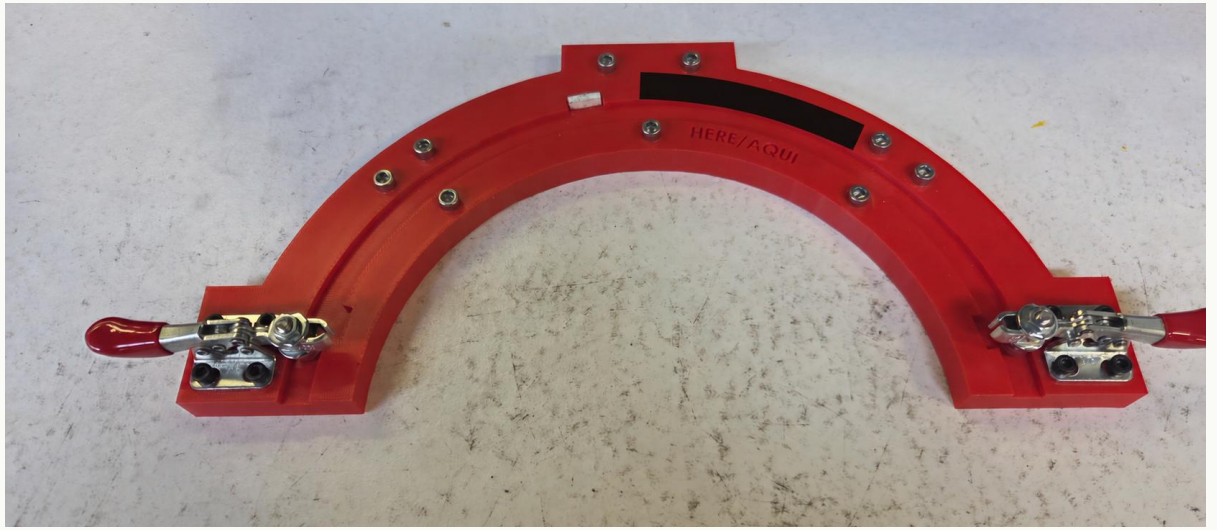
- Quick-change fixturing, designed with GD&T and DFM
- Fast, unambiguous inspection

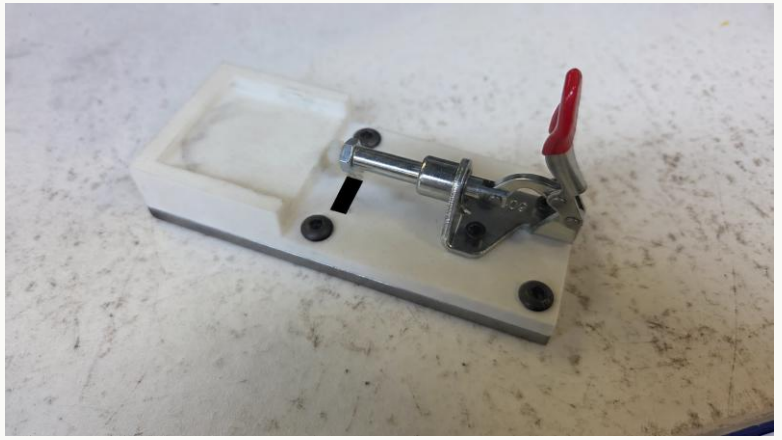
DECISION

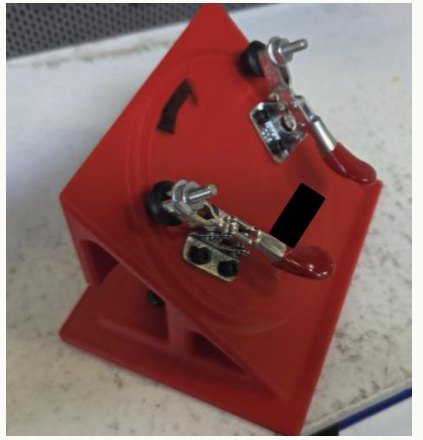
- 15 quick-change marking fixtures
- Go/No-Go gauges for inspection
- One-piece-flow changes

VERIFICATION

- Load/unload **6 s → 1 s** (83%); inspection **40 s → 2 s** (95%)
- Time between completed parts −55%







Marking fixtures. Identifiers removed. Go/No-Go gauges not pictured.

Qualifying an in-house cleaning process

Engineering Intern · Aug–Dec 2025 · Newburyport, MA

PROBLEM

- Precision cleaning of a Rolls-Royce component was outsourced, adding lead time, cleaning and shipping cost

CRITERIA

- Acceptable to Pursuit Quality and to Rolls-Royce
- Lower lead time and cost than the outside cleaner

DECISION

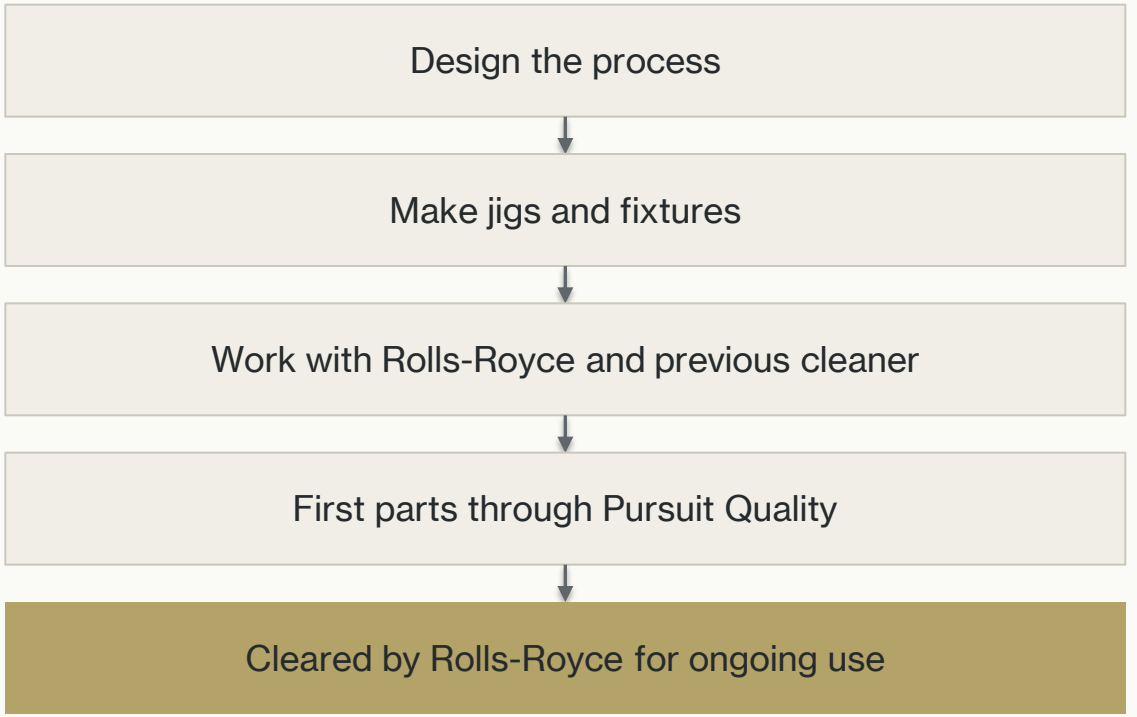
- Designed a new in-house cleaning process
- Made the jigs and fixtures for it
- Communicated with Rolls-Royce and the previous cleaner

VERIFICATION

- Took the first parts through Pursuit Quality
- **Cleared by Rolls-Royce** for ongoing use in Pursuit's shop

$1,000
saved per part

3 days
less lead time



Nathaniel Murcin

My work centers on mechanical and fluid systems design.

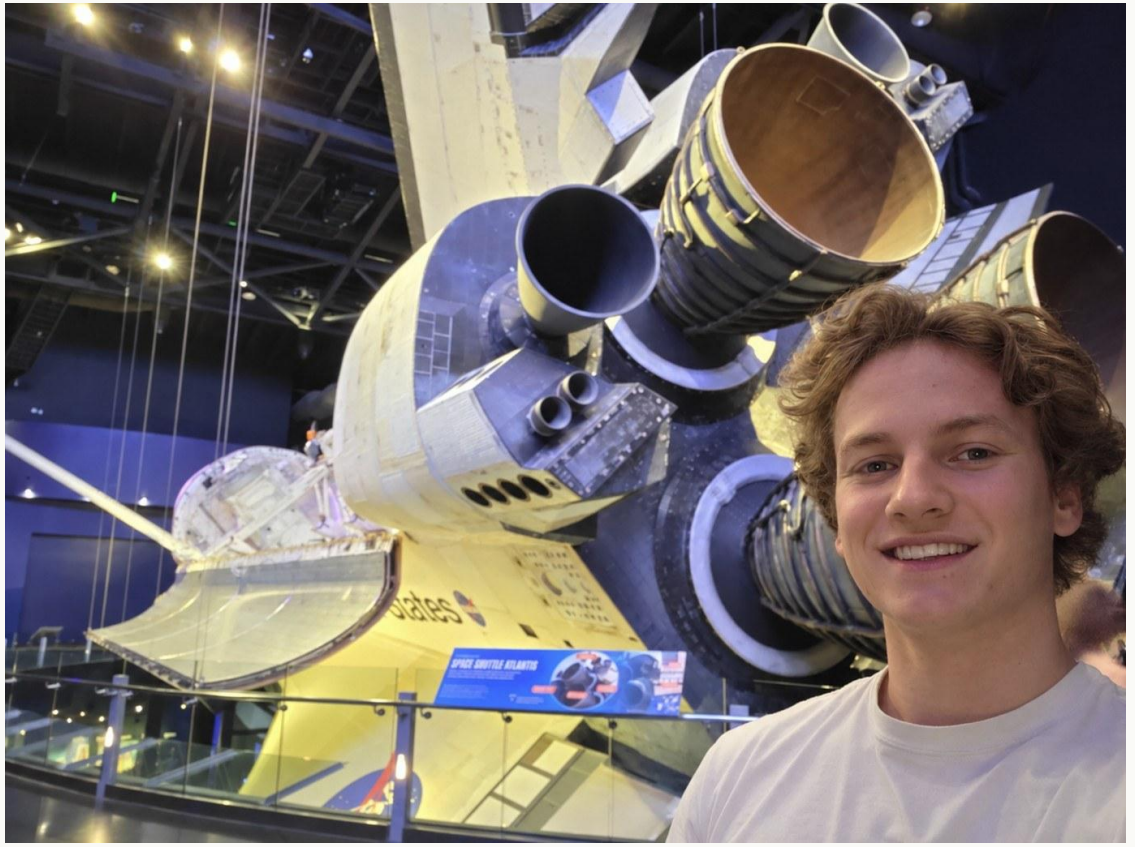
Aerospace Engineering · Georgia Institute of Technology

B.S. expected May 2027 · M.S. expected May 2028
Minor in Astrophysics · GPA 4.00/4.00

nmurcin3@gatech.edu

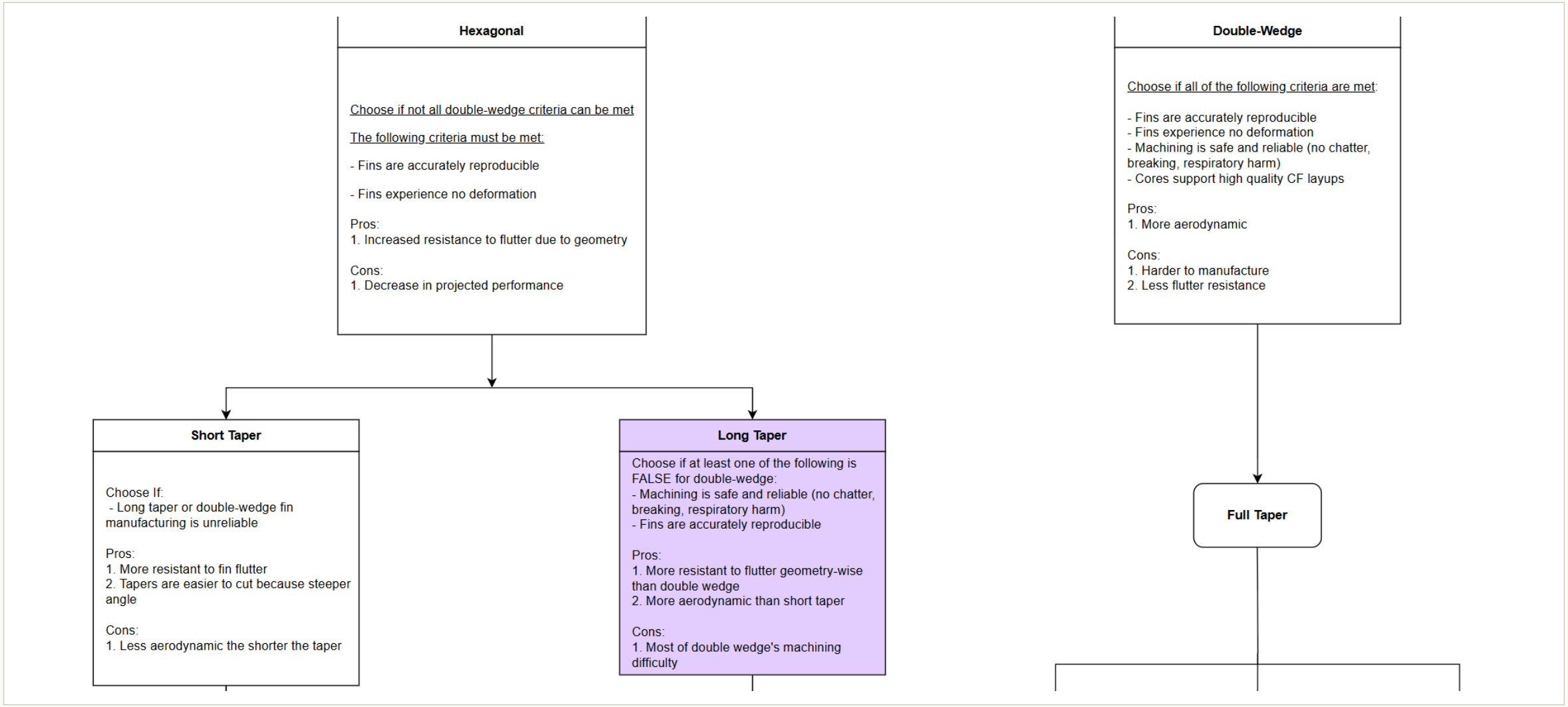
linkedin.com/in/nathaniel-murcin

nmurcin.github.io



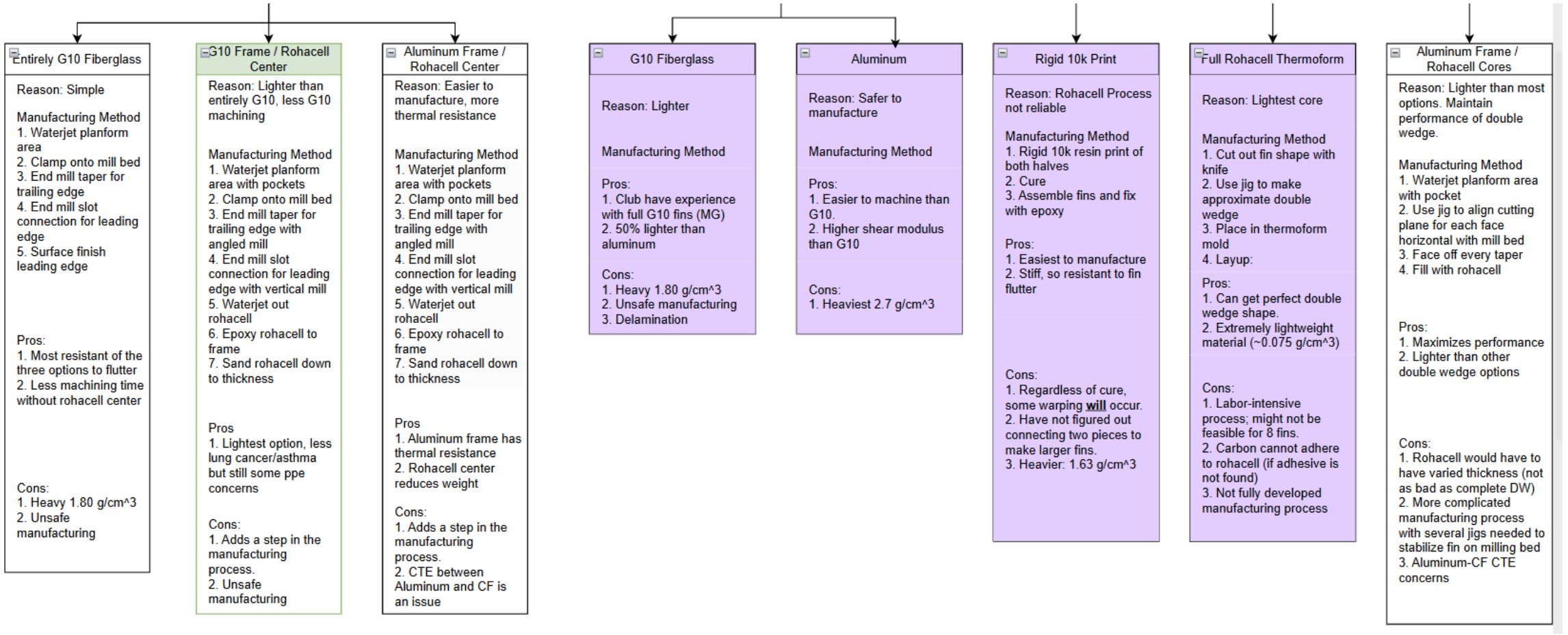
Space Shuttle Atlantis, Kennedy Space Center

Decision matrix: shape and taper



Purple boxes were eliminated. Zoom in for detail.

Decision matrix: core options



Purple boxes were eliminated. Zoom in for detail.